

Bitcoin, Blockchain, and Cryptocurrencies

MGT 4074 Fintech & Crypto Tokens, Fall 2025

Joseph Hall

Georgia Institute of Technology

September 10, 2025

Welcome

For Monday (9/15):
Read the Bitcoin Whitepaper:
<https://bitcoin.org/bitcoin.pdf>

Please create a MetaMask Wallet and fill out your **public** key in the Canvas quiz.
<https://support.metamask.io/start/getting-started-with-metamask/>

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D

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Forbes:

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The project's original vision was a two-tiered structure. The first monetary layer was to be an environment exclusively for transactions among authorized participants - specifically regulated financial institutions running nodes on the Central Bank-controlled network. The second would be tokenized bank deposits issued by regulated institutions to end customers.

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TL;DR: Brazil has decided not to record transactions of the “Digital Real” onto a blockchain. Instead, it will utilize a central database.

Roadmap

- 1 In the News: Brazil Abandons Blockchain for CBDC
- 2 **Math**
 - Hash Functions
 - Digital Signatures
 - Merkle Trees
- 3 **Blockchains**
 - A Centralized Blockchain
 - A Decentralized Blockchain
 - Mining
 - Some Basic Game Theory
 - Difficulty Adjustments
- 4 **Bitcoin**
 - History
 - Technical Details
 - User Experiences

Be Not Afraid

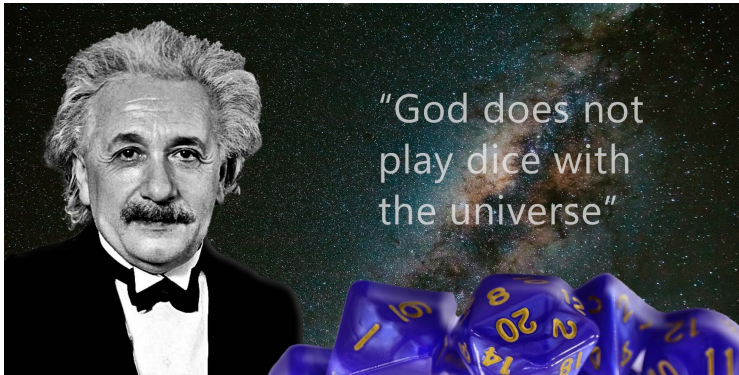
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 - Hash functions
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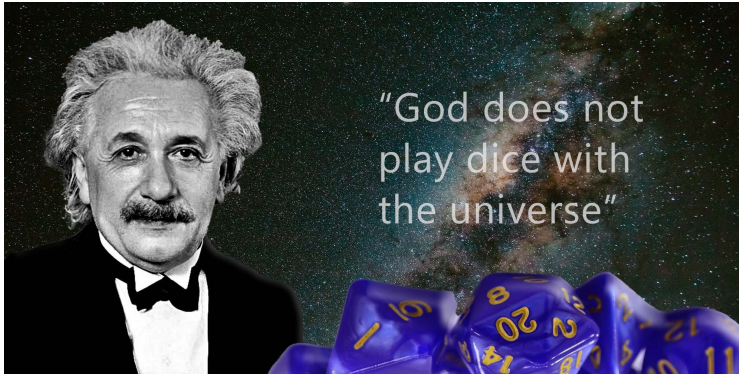
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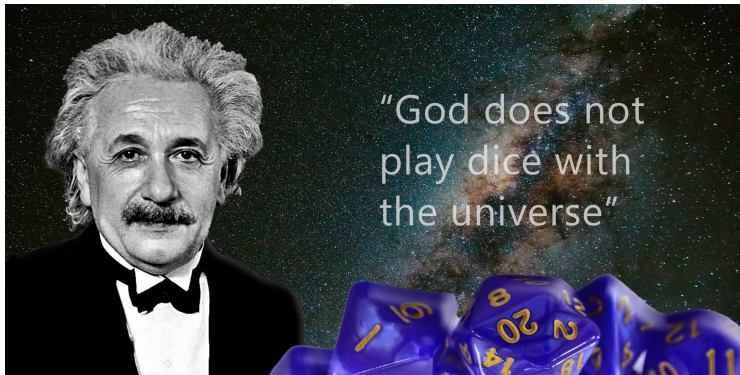
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 - The better you understand the math, the better you'll understand what is and is not technologically possible
 - You might be better than me at math





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- Anyone heard of the "Three Body Problem"?

Chaos

- Nature (and math) has many processes that are **chaotic**: deterministic, but unpredictable
 - Three-body problem
 - Physical dice, coins
 - Which numbers are prime?
- “Deterministically” driven by physics laws, but very sensitive to initial conditions, so from our perspective, as good as random

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 - We can simulate these (but always *some* error)

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A Chaotic Process



A Chaotic Process

Why does Cloudflare use lava lamps to help with encryption?

Randomness is extremely important for secure encryption. Each new key that a computer uses to encrypt data must be truly random, so that an attacker won't be able to figure out the key and decrypt the data. However, computers are designed to provide predictable, logical outputs based on a given input. They aren't designed to produce the random data needed for creating unpredictable encryption keys.

To produce the unpredictable, chaotic data necessary for strong encryption, a computer must have a source of random data. The "real world" turns out to be a great source for randomness, because events in the physical world are unpredictable.

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 - Convert text to binary numbers, 1101010...
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- Tl;dr, record your kid doing random stuff, and repeat it a lot of times

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A cryptographic hash function is a mapping $H : M \rightarrow D$ from a set M of possible messages of any length to a set D of digests, for example the set of strings of 0s and 1s of a fixed length, such as 256. See [Ramzan \(2012\)](#).

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In **digital signature schemes** (as we will see), a “digest” is a **signed message**.

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Answer: No, hash functions must be hard to invert.

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 - Bitcoins are mined by brute-forcing hashes to find x s that produce $H(x)$ with leading zeroes.

Modular Arithmetic

Powers of 3	3^1	3^2	3^3	3^4	3^5	3^6	3^7	3^8	3^9
Value	3	9	27	81	243	729	2187	6561	19683
Closest multiple of 7	0	7	21	77	238	728	2184	6559	19677
Remainder div. 7	3	2	6	4	5	1	3	2	6

- We use $x \bmod y$ to mean, “remainder of x when divided by y ”, like:

$$81 \bmod 7 = 4$$

- In python, $x \bmod y$ is just $x \% y$
- In R, it's $x \%\% y$

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 - But, using the public key, can't generate new valid signatures

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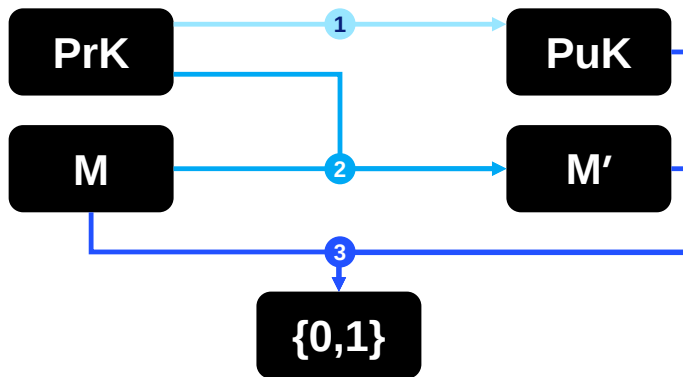
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In Bitcoin, these “messages” encode a list of transfers of blocks from one owner to another!

Digital Signature Schemes, Visually



Ok, but how do these signatures actually work?

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The easiest to implement from scratch, that I'm aware of, is called [ElGamal](#).

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Welcome! Today is Monday 9/15

There is a reading quiz today on the Bitcoin Whitepaper.

Midterm 1 will be discussed at the start of class. Your copies will be distributed at the end of class. *If you think you may appeal your score*, you must inform me at that time so I can photograph your exam and ensure you do not alter it.

Homework 1 has been posted to Canvas. As of today you will know everything you need to know to complete it. It is due at the start of class 7 days from today.

Midterm 1 Post-Mortem

Short Answer Question 1: The No-Trade Theorem.

Assumptions:

- Heterogeneous information (in no particular direction!)
- Common preferences (everyone values the same cash flows the same)
- (There is also an assumption of rational expectations, but that was not in the slides nor necessary to state for full credit on the exam)

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The **result of those assumptions** is that no trade should take place!

The intuition is that being offered a trade is sufficient reason to reject it, if there are no positive-sum trades!

The take-away is that trade in financial markets **must** be based on differences in preferences: e.g. someone who needs money today trades with someone who needs money tomorrow.

Midterm 1 Post-Mortem

Short Answer Question 2: *Seignorage* is the profit from printing money.

- **Specie** is a naturally-scarce material.
 - Seignorage can be earned through **coinage** and increased through **debasement**
- **Fiat** is a currency on which the government enforces coordination.
 - Seignorage is *maximized* with this form of currency
- **Crypto** is a currency secured with cryptography
 - The most valuable cryptocurrency, **Bitcoin**, has no seignorage
 - A cryptocurrency can be designed to give seignorage to its creator (this was not necessary to state in order to earn full credit).

In the News: Stablecoin Price War

From Bloomberg's Matt Levine:

I have always thought that issuing stablecoins is a great business. The way it works is:

1. People give you dollars in exchange for cryptographic tokens that you make up.
2. You invest their dollars in safe short-term assets, like a government money market fund yielding about 4%.
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It's like banking, but without interest or branches or credit risk: Instead of taking people's money and using it to make loans, you take their money and park it in Treasury bills and keep the interest for yourself. Your costs are low and your revenue is high.

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A: If deposits flow out of banks and into stablecoins, we will lose the liquidity-transforming, loan-supplying function of banks!

In the News: Stablecoin Price War

So where does this leave us?

That is controversial. The Financial Times reported last month that “banking lobbies including the American Bankers Association, the Bank Policy Institute and the Consumer Bankers Association last week warned lawmakers of a ‘loophole’ in regulation that will let some crypto exchanges indirectly pay interest to stablecoin holders,” which might “spark mass deposit outflows ... if customers choose to earn yield by holding stablecoins at crypto exchanges rather than coins or cash dollars at banks.”

But it also means that issuing stablecoins is not as great a business as I once thought. It was, when there were only a few stablecoins and the stablecoins acquired their network effects naturally. But in the long run, to be a competitive stablecoin, you might have to pay most or all of your interest income to crypto exchanges, and the exchanges might have to pass most or all of it back to the customers. “You hold people’s dollars for them, you collect interest, and you don’t give them any of it” really is a good business model, but it’s too good to last.

In the News: Stablecoin Price War

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Uses of Merkle Trees

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Merkle Trees are used in many applications to solve problems such as:

- Maintaining integrity of files stored on Dropbox.com, or *file outsourcing*
- Proving a transaction between Alice and Bob occurred, or *set membership*
- Transmitting files over unreliable channels, or *anti-entropy*

Construciton of Merkle Trees

Start with, say, 8 files ($f_1 \dots f_8$), and a hash function $H(\cdot)$

Construciton of Merkle Trees

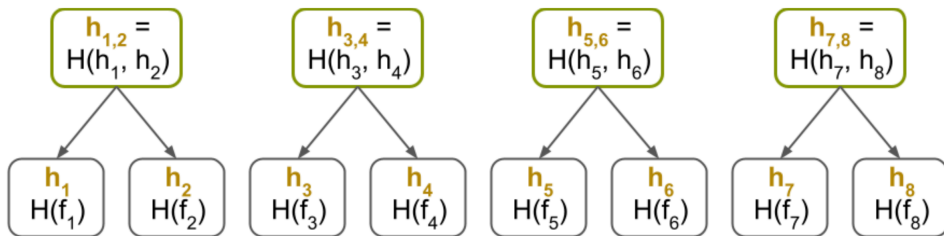
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Start by hashing each file $H(f_i)$:

$h_1 =$ $H(f_1)$	$h_2 =$ $H(f_2)$	$h_3 =$ $H(f_3)$	$h_4 =$ $H(f_4)$	$h_5 =$ $H(f_5)$	$h_6 =$ $H(f_6)$	$h_7 =$ $H(f_7)$	$h_8 =$ $H(f_8)$
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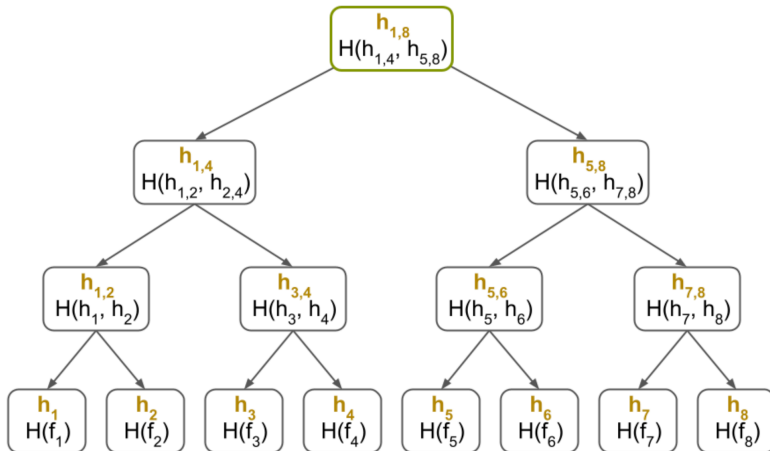
Construction of Merkle Trees

Then, create the second layer of the tree by hashing adjacent nodes together (this can be as simple as concatenating them!)



Construction of Merkle Trees

Continue until you have created an entire “tree”:



Construction of Merkle Trees

Question: Why is this tree upside-down?

Construction of Merkle Trees

Question: Why is this tree upside-down?

Answer: Because Computer Scientists have never gone outside, a necessary prerequisite to see a real tree

Verification with Merkle Trees

Upload your files ($f_1 \dots f_8$) **and** your Merkle Tree to Dropbox, storing only the **Merkle Root** $h_{1,8}$ locally.

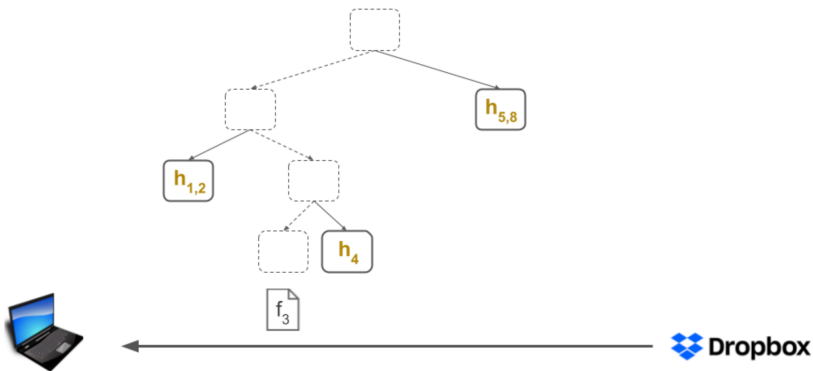
Verification with Merkle Trees

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Later, ask for the file you want, say f_3 , but make sure to also ask for the **Merkle Proof**: the hashes needed to compute $h_{1,8}$ from the desired **Merkle Leaf** (f_3)

Verification with Merkle Trees

The **Merkle Proof** lets you verify that you downloaded the real, original f_3 :



Advantages of Merkle Trees

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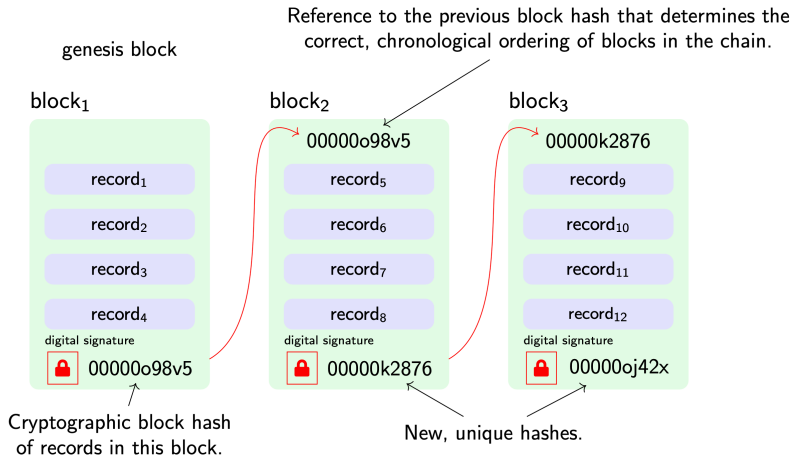
Answer: The cryptographic property of **Non-Invertability** ensures that it is prohibitively difficult to create a false Merkle proof.

Roadmap

- 1 In the News: Brazil Abandons Blockchain for CBDC
- 2 Math
 - Hash Functions
 - Digital Signatures
 - Merkle Trees
- 3 **Blockchains**
 - A Centralized Blockchain
 - A Decentralized Blockchain
 - Mining
 - Some Basic Game Theory
 - Difficulty Adjustments
- 4 Bitcoin
 - History
 - Technical Details
 - User Experiences

What is a Blockchain?

A **Blockchain** is a list of records that are linked together by hash functions



A Centralized Blockchain

Say that I, JPH, own and operate a private server which records a Blockchain. I don't know the users' private keys, but I control everything that is written on the blockchain. Can I:

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- Selectively cancel three messages ago, but not the last two?

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Centralized Blockchains are in fact widely-used:

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- Ripple XRP

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- Centralized Blockchains add an extra level of security: create a transaction history that can't be arbitrarily tampered with – only rolled back entirely or stopped pre-emptively.
- That's still a lot of power in the admin's hands!

Can we take away even more power from the admin? Can we make a set of rules to agree on a shared history of transactions without needing to trust anybody?

A Decentralized Blockchain

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- Bitcoin solves this through **proof of work**

Hash Puzzles

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 - 1 Hard to do
 - 2 Proves that you did something hard to do
- With puzzle solution in hand, you become the admin!
- **Incentives:** solving the puzzle gives you some BTC as a block reward
- **Consensus:** everyone works on the longest chain

-

Turning Back Time...

- Anyone can try, since anyone can try to mine and become admin
- When would I succeed?

Turning Back Time...

- Anyone can try, since anyone can try to mine and become admin
- When would I succeed?
 - Because everyone uses the **longest chain**, I must be able to hash faster than **everyone else combined**
 - Control 51% of the computing power in the Bitcoin ecosystem!
- For more details, see [Stanford CS251 slides](#)

Welcome! Today is Wednesday 9/17

Your first homework is due on Canvas by the **start of class** on Monday.

Also for **Monday**: read the Ethereum Whitepaper:

<https://ethereum.org/whitepaper/>

Guide your reading by thinking about the following questions:

- What is a Smart Contract?
- How do fees in Ethereum differ from fees in Bitcoin?
- Why do Turing-complete smart contracts require a different fee structure?

In the News: Google's Agent Payment Protocol

How it works: Establishing trust via mandates and verifiable credentials

AP2 builds trust by using Mandates—tamper-proof, cryptographically-signed digital contracts that serve as verifiable proof of a user's instructions. These mandates are signed by verifiable credentials (VCs) and act as the foundational evidence for every transaction.

Mandates address the two primary ways a user will shop with an agent:

- **Real-time purchases (human present):** When you ask an agent, “Find me new white running shoes,” your request is captured in an initial **Intent Mandate**. This provides the auditable context for the entire interaction in a transaction process. After the agent presents a cart with the shoes you want, your approval signs a **Cart Mandate**. This is a critical step that creates a secure, unchangeable record of the exact items and price, ensuring what you see is what you pay for.
- **Delegated tasks (human not present):** When you delegate a task like, “Buy concert tickets the moment they go on sale,” you sign a detailed **Intent Mandate** upfront. This mandate specifies the rules of engagement—price limits, timing, and other conditions. It serves as verifiable, pre-authorized proof that can allow the agent to automatically generate a **Cart Mandate** on your behalf once your precise conditions are met.

In both scenarios, this chain of evidence culminates in securely linking your payment method to the verified contents of the Cart Mandate. This complete sequence—from intent, to cart, to payment—creates a non-repudiable audit trail that answers the critical questions of authorization and authenticity, providing a clear foundation for accountability.

The Money Pot Game

- Two options:
 - ① Get \$1 for sure
 - ② Fight for a \$10 pot, which is equally divided between everyone who picks this option

The Money Pot Game

- Two options:
 - ① Get \$1 for sure
 - ② Fight for a \$10 pot, which is equally divided between everyone who picks this option
- Which do you pick?

The Money Pot Game

- What is going on?
 - Everyone wants the money pot
 - But the more people take the money pot, the less money there is for everyone
 - In “Nash equilibrium”, the money pot becomes no better than the “outside option”
- Similar settings: what’s the equilibrium condition?
 - Firms competing for profits?
 - Going to the beach on a sunny day?
 - Getting a table for Brunch on Sunday at 12pm?
- “Equilibrium” behavior is easy to predict in such settings
 - In equilibrium, the “good” deal is so crowded that it becomes just as bad as the “bad” deal

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- Game theory language: call these congestion games, or wars of attrition, or tragedy of the commons

BTC mining

- Suppose mining a block pays 0.001 BTC
- Suppose mining block costs \$60 in electricity
 - If BTC price $> \$60k$, everyone mines; blocks produced quickly
 - If BTC price $< \$60k$, nobody mines; tx's aren't included and the blockchain stops working!
- The fact that mining rate depends on BTC vs electricity price is not desirable...
- Solution: use difficulty adjustment to maintain constant block production rate

BTC Difficulty Adjustment

- Blocks should be produced every 10 minutes
- Every 2016 blocks (approx 14 days), Bitcoin considers average block production rate, and...
 - Increases/decreases hash problem difficulty, if block production is too fast/slow
- What determines how fast blocks are produced, fixing hash rate?

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- What determines how many people mine?
 - How profitable mining is, that is, how high BTC price is, vs electricity price

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What happens to mining when BTC price goes up?

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- More people try to mine
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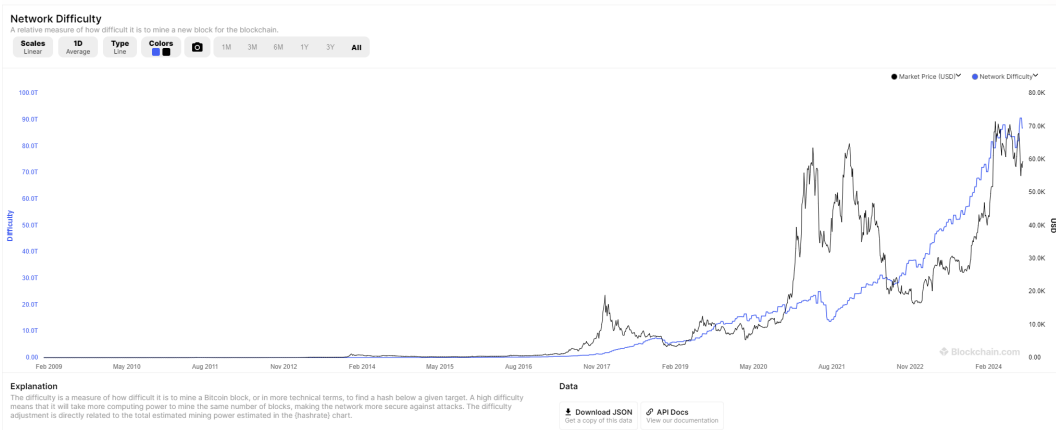
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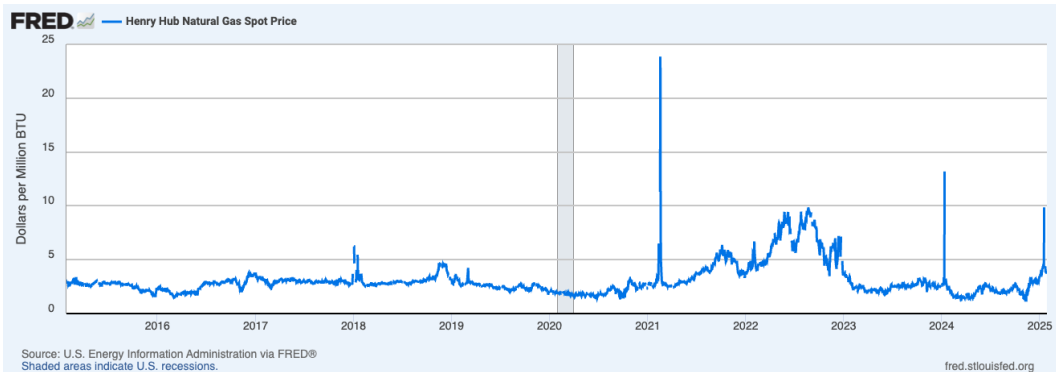
Prediction: difficulty should increase with BTC price

Difficulty vs BTC price



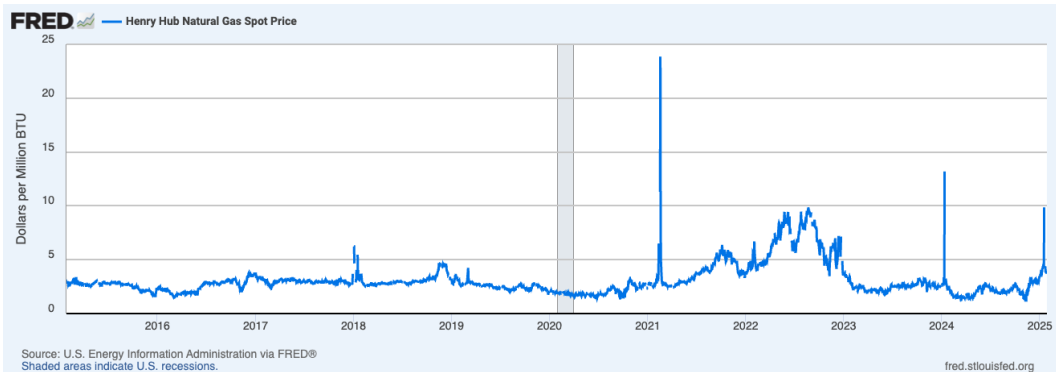
Source: [Blockchain.com](https://blockchain.com)

Energy Prices...



Source: [FRED](#)

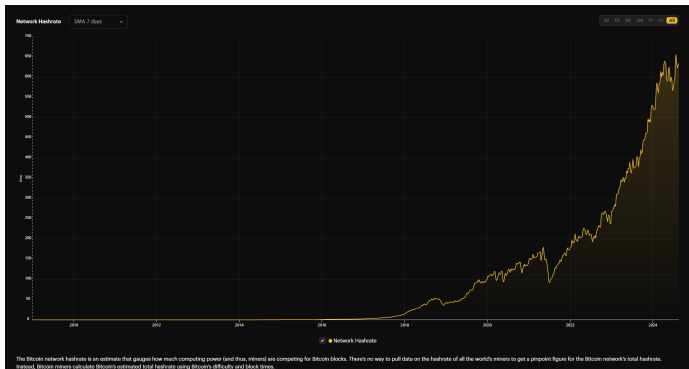
Energy Prices...



Source: [FRED](#)

A **BTU** is a “British Thermal Unit”, the energy required to heat a pound of water by 1 degree Fahrenheit (about one match)

Technological Improvement



- 842 Exahashes/s = 842 billion billion hashes a second!
- Each living person in the world could shake each other living person's hand 10 times, every second!
- ~ 1% of global energy consumption...

Back-of-Envelope Calculation

As of 2024,

- 164,250 BTC mined per year, worth around \$8bil
 - In theory, what should total mining costs be?

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- 164,250 BTC mined per year, worth around \$8bil
 - In theory, what should total mining costs be?
- Back of envelope calculation...
 - World electricity consumption approx 25,000TWh; BTC mining around 0.5%, so 125TWh
 - At US price \$0.12/kWh, this is approx \$15bil!
 - (a kWh (kilo-Watt-hour) is 3,412 BTUs)
- Electricity is probably somewhat cheaper, getting us close to \$8bil

1. *Journal of Management Studies*, 1990, 27, 1, 1-14.

- Approximately every 4 years, issuance of BTC drops by half
- Last halving: April 2024, now 3.125 BTC per block
- Does this affect block production rate?
 - No, difficulty adjustment

BTC Halving

- Approximately every 4 years, issuance of BTC drops by half
- Last halving: April 2024, now 3.125 BTC per block
- Does this affect block production rate?
 - No, difficulty adjustment
- Also implies there will only ever be 21 million BTC in existence!
 - By approximately 2140, the block reward will drop to less than the smallest unit of bitcoin ($1/10^8$) and become equal to zero.
 - Difficulty adjustment ensures that block production rate is approximately constant

Why 21 Million?

Satoshi: "My choice for the number of coins and distribution schedule was an educated guess. It was a difficult choice, because once the network is going it's locked in and we're stuck with it. I wanted to pick something that would make prices similar to existing currencies, but without knowing the future, that's very hard. I ended up picking something in the middle. If Bitcoin remains a small niche, it'll be worth less per unit than existing currencies. If you imagine it being used for some fraction of world commerce, then there's only going to be 21 million coins for the whole world, so it would be worth much more per unit. Values are 64-bit integers with 8 decimal places, so 1 coin is represented internally as 100000000. There's plenty of granularity if typical prices become small. For example, if 0.001 is worth 1 Euro, then it might be easier to change where the decimal point is displayed, so if you had 1 Bitcoin it's now displayed as 1000, and 0.001 is displayed as 1."

Transaction Fees

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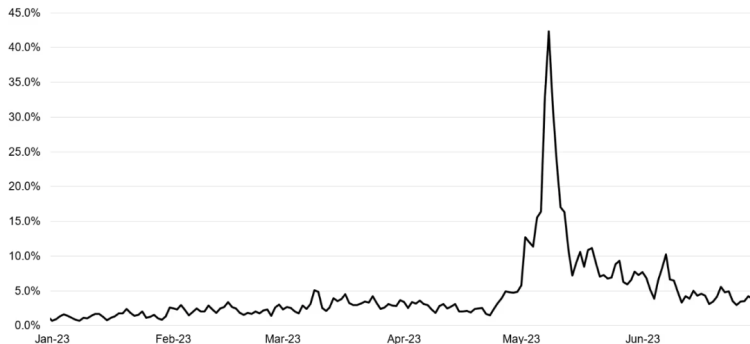
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- As of 2/4/25, the going rate to complete your transaction within 30 minutes is 3 sat/vB, or \$0.42 for an average transaction
- A “**mempool**” is a pool of transactions waiting to be added to the bitcoin blockchain

Explore mempools at <https://mempool.space/>

Transaction Fees as % of Block Revenue

Daily Transaction Fees % of Total Block Reward

Source: Galaxy Research



Data: Coin Metrics

Incentives

- A centralized authority can “censor”, i.e. not include tx’s
- Decentralization and competition “ensure” performance
 - If you don’t include my tx, the next miner will
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- Incentives not that strong: tx fees $< 10\%$ of mining revenues
 - But, it’s free \$\$!

Further Reading on Mining Industry

- Galaxy digital 2023 and 2024
- Cool paper on mining pools

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Satoshi Nakamoto

- Started sending emails about Bitcoin August 2008
- Oct 2008: Bitcoin [white paper](#), code released
- January 9: first block mined!
 - “The Times 03/Jan/2009 Chancellor on brink of second bailout for banks”
- May 22, 2010: Laszlo Hanyecz bought two Papa John’s pizzas for 10,000 BTC, now worth \$600mil USD!
- Quickly became popular among black markets, [Silk Road](#)...
- Founder Ross Ulbricht was recently pardoned by Trump

Bitcoin Prices



Source: [CoinMarketCap](#)

Each BTC Block is a Merkle Tree of Transactions!

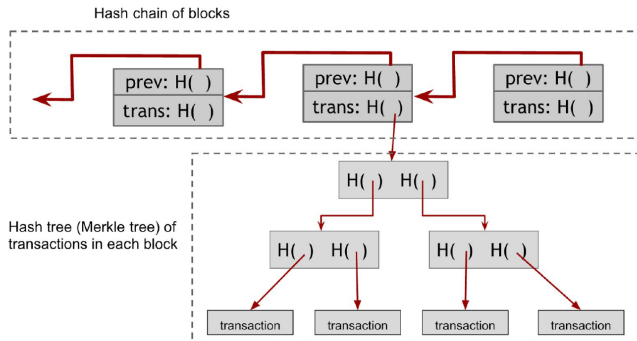


Figure 3.8. The Bitcoin block chain contains two different hash structures. The first is a hash chain of blocks that links the different blocks to one another. The second is internal to each block and is a Merkle Tree of transactions within the blocks.

- Valid block headers' hashes start with a big number of 0's (mining puzzle)

Mining, in Practice

- Miners connected to each other in a network
- If you find a valid block, tell all other miners about it
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 - Consensus: if everyone else uses "longest chain", I do better by accepting your win, and trying to mine the next block

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- What if I try to spend twice?

Fees and Times

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 - Tx's deeper down the blockchain are more "final" (harder to "double-spend")
- Long confirmation times a little awkward for "merchant" tx's

User Experience

- In practice, private keys held using “wallet” software or hardware
 - Software: Exodus, Electrum, MetaMask...
 - Hardware: Ledger, Trezor...
 - Some mobile app wallets, which I’m not familiar with
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 - I use MetaMask
- In theory, no credit risk – you’re holding your own BTC!
 - As long as internet + Bitcoin exists, you can use your BTC!

Don't Lose Your Keys...

As It Happens

This man owns \$321M in bitcoin — but he can't access it because he lost his password

Stefan Thomas has 2 guesses left before he's locked out of his fortune forever

CBC Radio - Posted: Jan 15, 2021 5:02 PM CST | Last Updated: January 15, 2021



Stefan Thomas, the founder and CEO of micropayment streaming service Coil, has \$321 million worth of bitcoins he can't access. (Submitted by Stefan Thomas)



284 comments

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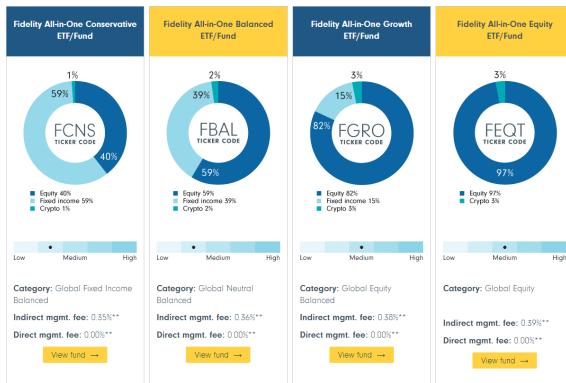
“Buy the Market”

The CAPM says buy the market – and BTC's market cap is now approx 2% S&P 500!

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Build your portfolio with confidence.



The “Market Cap” Illusion

Market Capitalization is just units outstanding times unit price

- Say everyone tried to sell their BTC tomorrow. Would they get that price for it?

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- Say everyone tried to sell their BTC tomorrow. Would they get that price for it?
- Traditional assets (stocks, bonds) generate **dividends** or **coupons**
 - If a major owner of a stock worth 2% of the S&P500 sold, the supply would be absorbed relatively quickly
 - Investors are willing to substitute frictionlessly between stocks if one is temporarily undervalued

